

Computing Ideas Hardware



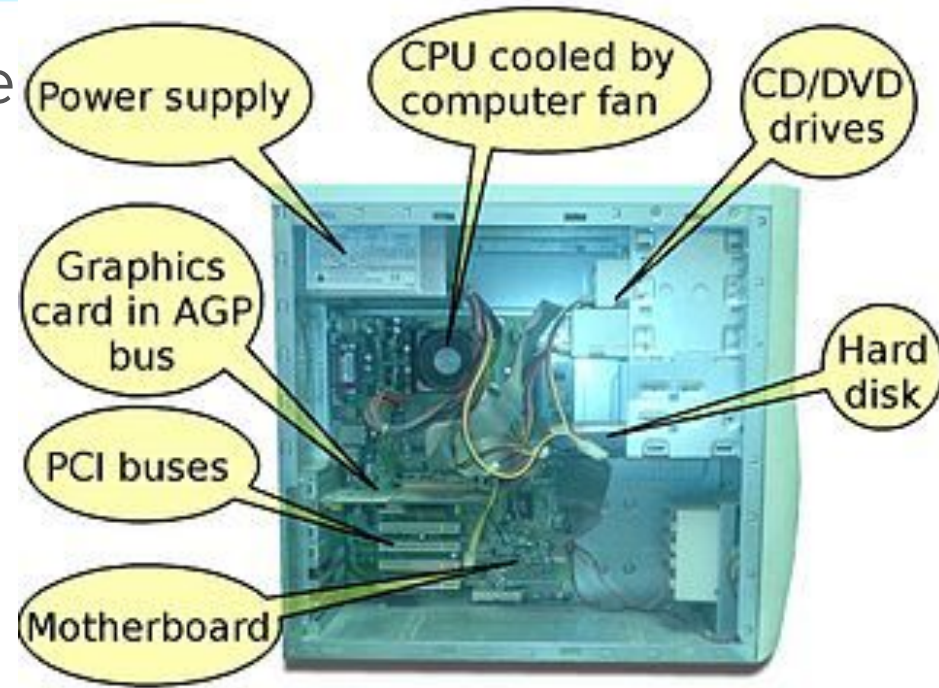
CodeHS

What is hardware?

The physical components of your computer.

Hardware

- The physical parts that make up your computer
- Uses electrical signals to carry out instructions
 - Electric signals implement binary, the language of the computer



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How does hardware work?

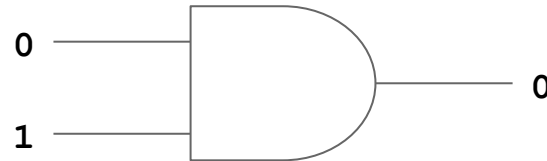
The basic overview
of how hardware
operates.

Bits, Bytes, Binary

- Binary is a number system with only 1's and 0's
- Computers understand instructions in binary
- A single binary digit is a bit: 0 or 1
- Eight bits is a byte: 1101 1100
- A byte is a unit of digital data
 - 1 byte = 1 letter
 - 1 Megabyte = a short song

It's All Logic

- Hardware built using logic gates
 - Physical components that process incoming binary data and emit electrical signals
- All computations are performed only using logical operations



Main Hardware Components

Every computer has at least these three parts.

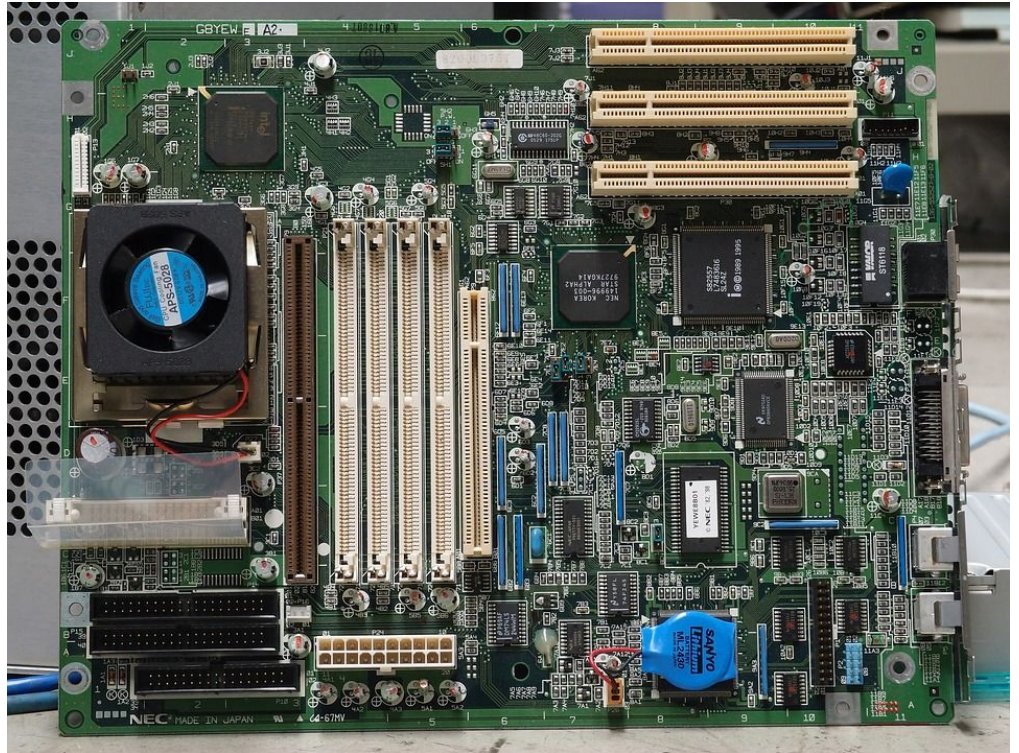
Three Main Components

- Memory
- Motherboard
- Central Processing Unit (CPU or processor)



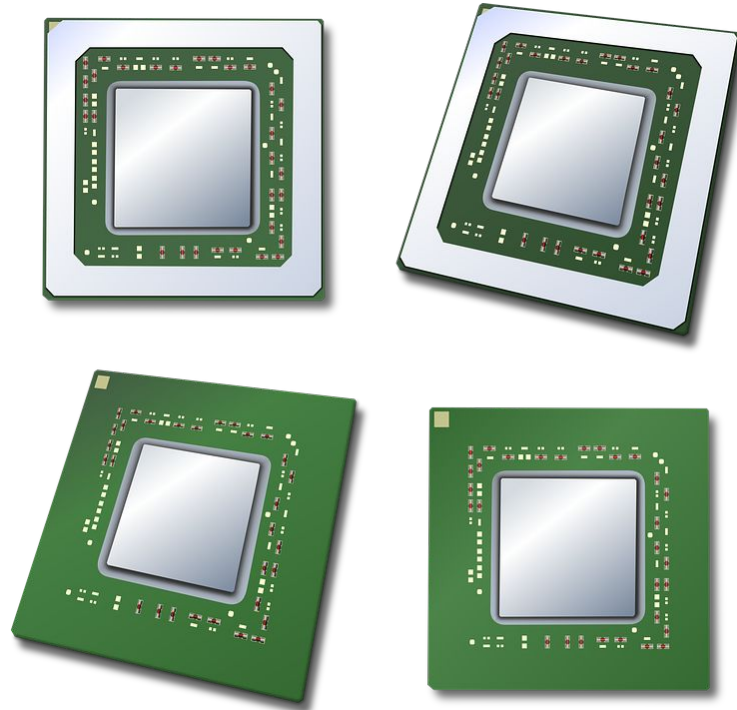
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Four CPUs

Memory

- Main Memory
 - Stores currently executing program instructions
 - Stores data the computer needs in order to work
- External Memory
 - Memory computers use to store data long-term
 - Memory users use to store data

Main Memory

- Random Access Memory (RAM)
 - Will only store information if the power is on
 - OS loaded into RAM memory at start-up
 - Stores currently running applications and corresponding data
- Read Only Memory (ROM)
 - Will keep information stored regardless of if the power is on
 - Where BIOS is stored

External Memory

- Used to store data long term
- Will keep data regardless of whether or not it is being powered
- Examples
 - CDs
 - USB
 - Hard drive
 - Floppy disks

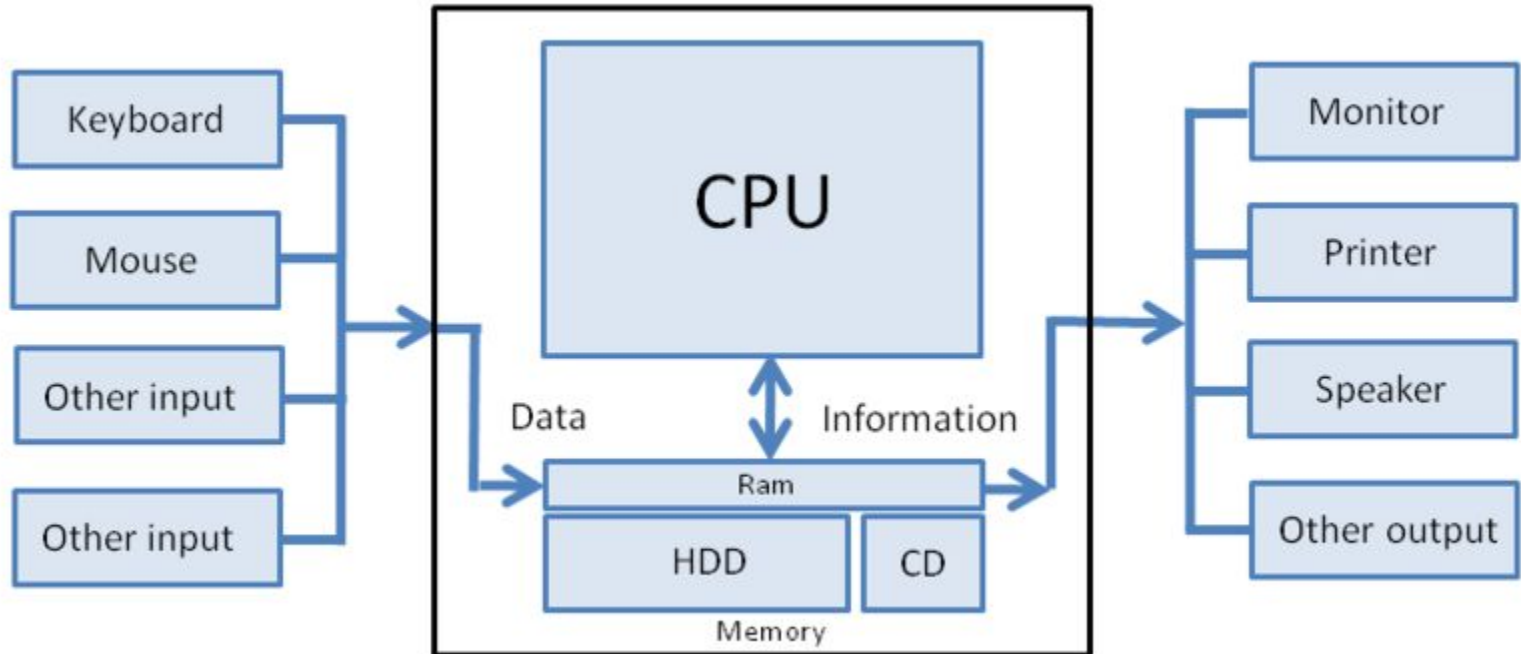
Motherboard

- Where the processor and memory live
- Connects the different parts of computer together
 - Input
 - Output
 - Storage
- Responsible for creating the clock signal for the CPU

CPU

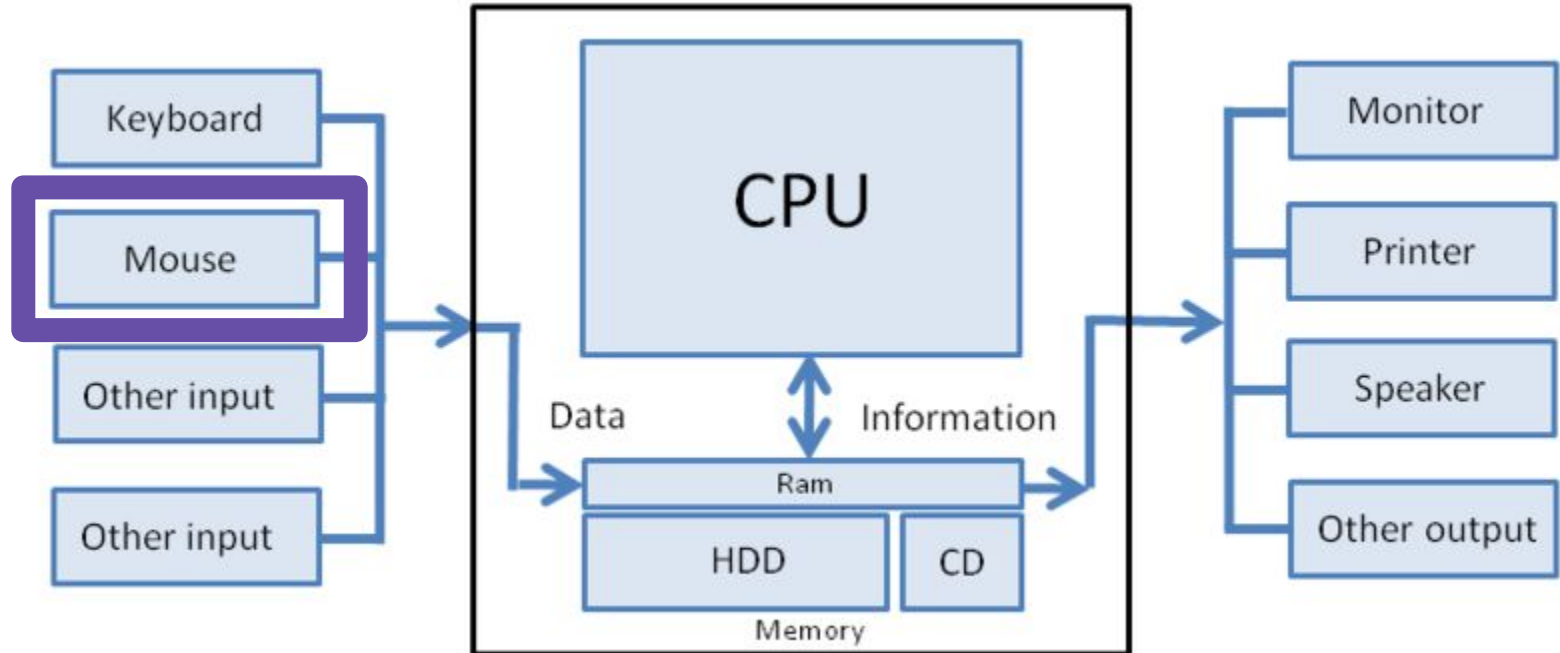
- Responsible for actually carrying out the instructions of a program
- Executes the instructions it receives from the computer's software
- Every instruction executed on a clock cycle
 - Every 'tick' of the clock pushes electricity through the computer to perform computations

Tying It All Together



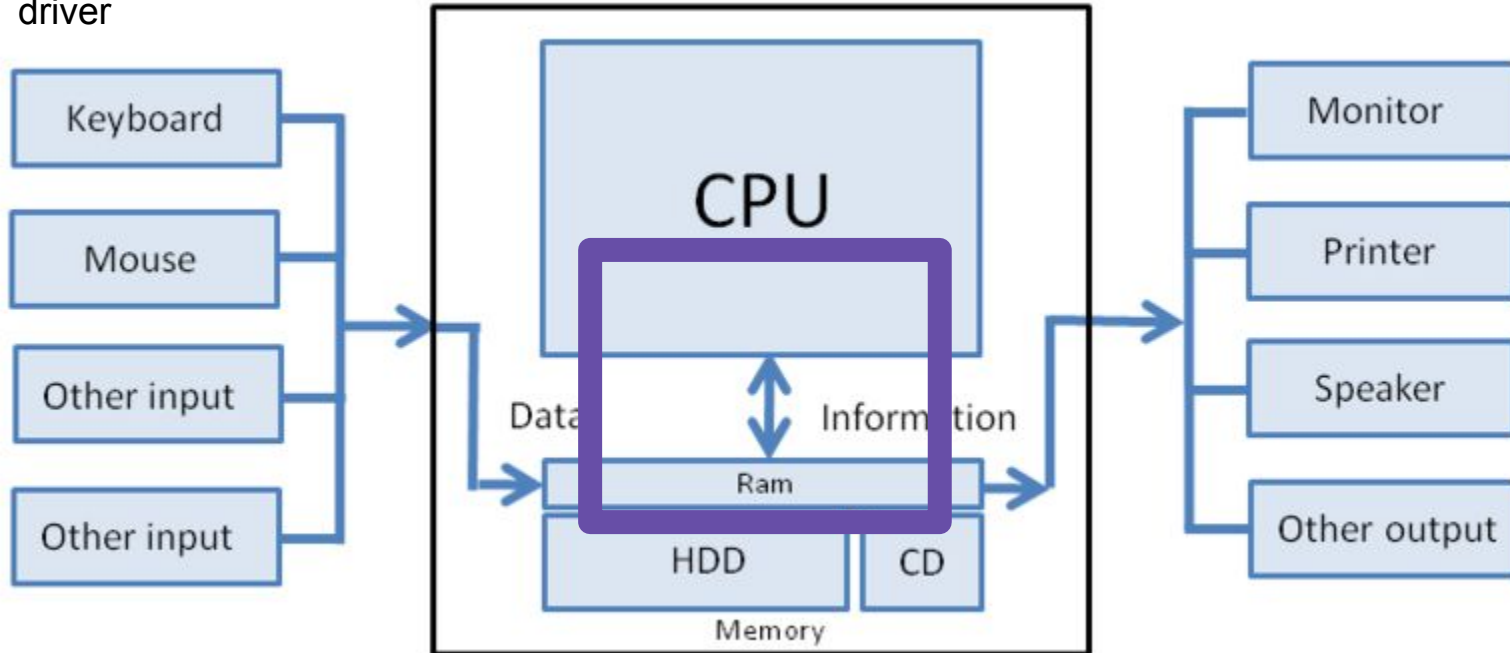
Tying It All Together

You give input to the computer by clicking on the “Print” choice in your text editor



Tying It All Together

The CPU takes the input, and performs some calculations. It gets the file from memory and prepares it to send to the printer. It gets the printer ready by accessing the device driver



Tying It All Together

The printer takes over and prints out the document.

